(Please write your Enrollment No. 01501042022)

End-Term Examination (Regular)
(CBCS)(SUBJECTIVE TYPE)(Online)
Course Name: B.Tech IT/MAE/DMAM, (Regular) Semester: Second
(July, 2023)

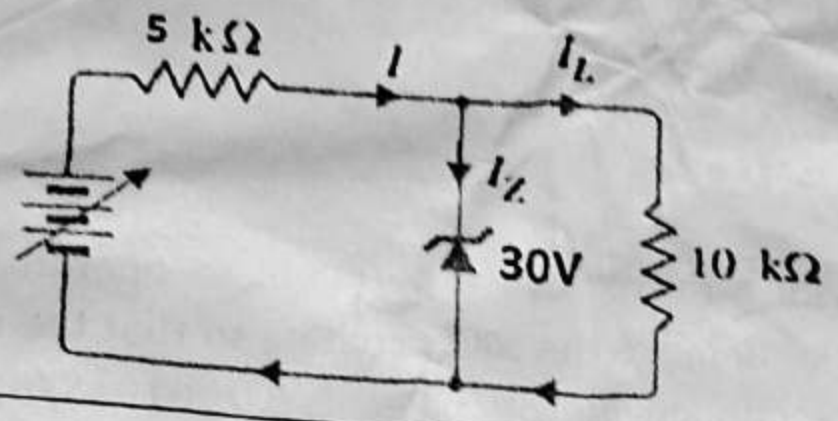
Subject Code: BAS-104	Subject: Applied Physics II
Time :3 Hours	Maximum Marks :60

Note: Q.1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Use the following:

Speed of Light, $c = 3 \times 10^8 \text{m/s}$
Charge of electron, $e = 1.6 \times 10^{-19} \text{C}$
Mass of electron, $m_e = 9.1 \times 10^{-31} \text{kg}$
Planck's Constant, $h = 6.626 \times 10^{-34} \text{m}^2\text{kg/s}$
Rydberg's Constant, $R = 1.09 \times 10^7 \text{m}^{-1}$
Boltzmann's Constant, $k_B = 1.38 \times 10^{-23} \text{m}^2\text{kgs}^{-1}\text{K}^{-1}$
Permeability of Free space, $\mu_o = 4\pi \times 10^{-7} \text{N/A}^2$
Permittivity of free space, $\varepsilon_o = 8.85 \times 10^{-12} m^{-3} kg^{-1} s^4 A^2$

Q1				
	(a) Define skin depth. To design a microwave experiment to operate at a frequency of 15GHz, how thick would you like to make the silver coating so that the microwaves do not penetrate the coating? The conductivity of silver is $6.0 \times 10^7 S/m$ and its relative permeability $\mu_r \approx 1$.	**(5*4=20)**		
	(b) (i) Determine the miller indices of a plane with intercepts $a/2$, $b/3$ and c along the x, y and z axes respectively. (ii) Sketch the following planes within a cubic unit cell: $(0\ \bar{1}\ \bar{1})$ $(1\bar{3}1)$ and (102)			
	(c) Two spaceships leave the earth in opposite directions, each with a speed of $0.5c$ with respect to the earth. What is the velocity of spaceship 1 relative to spaceship 2?			
	(d) Describe the importance of Moseley's work on X-rays. State and explain Moseley's law. Show it graphically.			
	UNIT-I			
Q2	*(a) Derive Maxwell's* equations in both differential and integral forms. Explain the physical significance of each equation. (b) A plane electromagnetic wave travelling in + x direction in an lossless dielectric medium of $\mu_r = 1$ and $\varepsilon_r = 3$ has a peak electric field intensity $\lvert\vec{E}\rvert =$ 6 V/m. Find (i) the velocity of the wave in the medium (ii) the intrinsic impedance of the medium (iii) the peak value of the magnetic intensity (iv) peak value of Poynting vector.	**(5,5)**		
Q3	(a) What is displacement current? Explain how Maxwell corrected and presented the Ampere's law in its generalized form. (b) In a dielectric ($\sigma = 10^{-4}$S/m, $\mu_r = 1$, $\varepsilon_r = 4.5$), the conduction current density is given $J_c = 0.4\cos(2\pi \times 10^8 t)\ \text{A/m}^2$. Determine the displacement current density.	**(5,5)**		
	UNIT-II			
Q4	(a) Using the band theory of solids, differentiate between the following: (i) n-type and p-type semiconductors (ii)Conductors, Semiconductors and Insulators (b) Calculate the density of holes and electrons at room temperature $(= 27^o C)$	**(5,5)**		

	of a pure Germanium crystal, if energy band gap is equal to $E_g = 0.7eV$.		
Q5	(a) Show that the Fermi level of p-type semiconductor lies near valence band. Draw the relevant diagrams. (b) How can we provide voltage stabilization using a Zener diode? Explain its working principle with suitable diagram. For the following circuit, find the (i) maximum and (ii) minimum values of Zener diode current 	(5,5)	
	UNIT-III		
Q6	(a)Using Lorentz transformation equations, derive expression for time dilation and length contraction. (b) Determine the length of a rod of length $10m$ in a frame of reference which is moving with $0.6c$ velocity in a direction making 30° angle with the rod.	(5,5)	
Q7	(a) Derive Lorentz transformation Equations for two inertial frames of references S and S', where the S' frame is moving with relative velocity v along $+x$ direction. Show that the Lorentz transformations are equivalent to Galilean transformations for $v \ll c$. (b) An event occurs in the frame of reference $S(x, y, z, t)$ at $x = 100m$, $y = 10m$, $z = 5m$ and $t = 10^{-4}$ seconds. Calculate the coordinates of this event when viewed from another reference frame $S'(x', y', z', t')$ which is moving along the $+x$ axis with relative velocity $v = 2.7 \times 10^8 m/s$ using both the (i) Galilean and the (ii) Lorentz transformations.	(5,5)	
	UNIT-IV		
Q8	(a) Derive Bragg's law of x-ray diffraction through crystals. A beam of X-rays of wavelength $0.08nm$ is diffracted by (111) crystal plane of rock salt with lattice constant equal to $0.28nm$. Find the diffraction angle for the second-order Bragg's diffraction. (b) What are continuous and characteristics X-rays and how are they produced? An X-ray tube operates at the voltage 40kV. Find the maximum speed of electrons striking the anti-cathode and shortest wavelength of X-rays produced.	(5,5)	
Q9	(a) What is transfer function? Briefly explain a typical signal response curve of a sensor including dead band, saturation, linear portion and errors. (b) The transfer function of a sensor is controlled by four parameters A_1, A_2, A_3 and A_4. The response R with respect to the stimulus k is defined as $R = \frac{A_1 + A_2 k}{A_3 + A_4 k}$. Find the expression for the sensitivity of this sensor.	(5,5)	